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CSE 13S Spring 2021
Assignment 2: A Small Numerical Library
Design Document

New changes in red

# Purpose

The purpose of this assignment is to create a small numerical library and test harness. The library will include arcsin, arccos, arctan, and log functions. I plan on using a Taylor series to implement arcsin, arccos, and arctan. I will be using Newton's method to approximate the logarithm function. The test harness will allow me to compare my results to the standard <math.h> library and create a table showing the values and differences. Arcsin and arccos will be tested in the range [−1,1) with steps of 0.1. Arctan and log will be tested in the range[1,10) with steps of 0.1.

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# Layout/Structure

- The code for the functions will be in mathlib.c. This will contain the arcSin, arcCos, arcTan, and Log functions.
- Mathlib-test.c contains the main() program and acts as a test harness for the math library.
- The getopt() command line options that mathlib-test.c supports are:
    - -a: to run all tests.
    - -s: to run arcsin tests.
    - -c: to run arccos tests.
    - -t: to run arctan tests.
    - -l: to run log tests.

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# Functionality

- The arcSin function will be implemented using the inverse method with Newton's method and binary search. Values close to -1 will be calculated with binary search, values close to 1 will be calculated with the inverse method, and other values will be calculated with Newton's method.
- The arcCos function will be implemented using the arcSin function. I will use the equation $\arccos(x) = \frac{\pi}{2} - \arcsin(x).$
- The arcTan function will be implemented using the arcSin function and the equation

$$\arctan(x) = \arcsin\left(\frac{x}{\sqrt{x^2+1}}\right)$$ for x>0.

- The Log function will be implemented using Newton's method by computing the inverse of $e^x$. The following equation will allow us to implement logarithms with base e:

$$x_{k+1} = x_k + \frac{y - e^{x_k}}{e^{x_k}}.$$

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# Pseudocode

```
Abs(x) function:
        Return (x < 0) ? -x : x

Exp(x) function: (from Piazza)
        term = 1, sum = 1
        For (k=1; Abs(term) > EPSILON; k++):
                term *= x/k
                sum += term
        Return sum

Sqrt(x) function: (from Piazza)
        y = 1.0
        assert (x >= 0)
        For (guess = 0.0; Abs(y-guess) > EPSILON; y = (y + x / y) / 2.0):
                Guess = y
        Return y
```

Removed Pow function because it was unnecessary

```
arcSin(x) function:
        k=0.0
        Prev = sin(k)
        If (x>0.99):
                For (guess = 0.0; Abs(k-guess) > EPSILON; k = (k + x (x-sin(k)) / cos(k))):
                        Guess = k
                Return k
        If (x < -0.99):

                Declare m, l= -π/2, h=π/2
                Do:
                        m=(l+h)/2.0
```

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                If (Sin(m) < x):
                        l=m
                Else:
                        h=m
            While (Abs(l - h) > EPSILON / 2)
            Return m
        Else:
            while(Abs(prev - x) > EPSILON)
                k=k + (x - prev) / cos(k)
            Return k

arcCos(x) function:
        Return π/2-arcSin(x)

arcTan(x) function:
        Return arcSin(x / Sqrt(x * x + 1))

Log(x) function:
        Declare guess
        k=1.0
        For (guess=0.0; Abs(k-guess) > EPSILON; k=k+(x-Exp(k))/Exp(k)):
            guess=k
        Return k
```

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# Draft Work

$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}$$

$$\sin^{-1}(y) = x$$

$$\sin(\sin^{-1}(y)) = \sin x$$

$$y = \sin x$$

$$y - \sin x = 0$$

$$x_{k+1} = x_k - \frac{y - \sin x_k}{-\cos x_k}$$

$$x_{k+1} = x_k + \frac{y - \sin x_k}{\cos x_k}$$